

Design of Photonic Switch based on $Ge_2Sb_2Te_5$ embedded SOI rib waveguides for operation at 1 THz

*Thesis submitted in partial fulfillment of the
requirements for the degree*

of

Bachelors of Technology in Engineering Physics

by

Yagyaansh Khaneja
17122031

Under the supervision of

Prof. Rajesh Kumar



PHYSICS DEPARTMENT

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

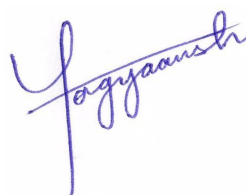
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Declaration of Authorship

DECLARATION

Project Title	Design of Photonic Switch based on $Ge_2Sb_2Te_5$ embedded SOI rib waveguides for operation at 1 THz
Author	Yagyaansh Khaneja
Enroll. No.	17122031
Supervisor	Prof. Rajesh Kumar

I declare that this Thesis entitled *Design of Photonic Switch based on $Ge_2Sb_2Te_5$ embedded SOI rib waveguides for operation at 1 THz* is the result of my own work except as cited in the references. The Thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.



Yagyaansh Khaneja

17122031

Physics Department
Indian Institute of Technology Roorkee



Physics Department
Indian Institute of Technology Roorkee
Roorkee-India

CERTIFICATE

This is to certify that the thesis entitled **Design of Photonic Switch based on $Ge_2Sb_2Te_5$ embedded SOI rib waveguides for operation at 1 THz**, submitted by **Yagyaansh Khaneja** (Student ID: 17122031) an undergraduate student of the **Physics Department** has been examined. Upon recommendation by the examination committee, we hereby accord our approval of it as the presented work and submitted report fulfill the requirements for its acceptance in partial fulfillment for the degree of *Bachelor of Technology in Engineering Physics*.

Rajesh Kumar

Physics Department
Indian Institute of Technology Roorkee

G. D. Varma
Professor and Head

Physics Department
Indian Institute of Technology Roorkee

Place: Roorkee

Date: 19 August 2021

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Indian Institute of Technology Roorkee

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Abstract

Design of Photonic Switch based on $Ge_2Sb_2Te_5$ embedded SOI rib waveguides for operation at 1 THz

by [Yagyaansh Khaneja](#)

It is necessary to miniaturize terahertz systems for a broad-scale application of Terahertz technology. Dielectric waveguides are a critical component in this regard. The contrasting and strikingly different optical properties of different phases of chalcogenide phase change materials (PCMs) have spurred the development of various novel photonic devices such as all-optical non-von Neumann memory, and nano pixel displays color rendering and reconfigurable nanoplasmonics. However, the exploration of chalcogenide photonics has traditionally been limited to optical and infrared frequencies. This work is a study (using simulations) of one of the many theoretical designs of an on-off switch based upon GST-embedded SOI rib waveguides for operation in the far-infrared frequency of 1 THz. We can change the state of the proposed switch by employing electrical or optical pulses, and no further electrical bias is needed to maintain a particular state. Because the active material (GST) has two "stable" phases, i.e., crystalline and amorphous, the device holds itself in either of the two states, and the voltage or optical pulse needs to be applied only during transition. To enhance the interaction of light with GST, the approach of embedding the active material GST within the Si waveguide has been taken. The dependence of the dimensions of the active region of the switch has been studied to achieve low insertion loss and high extinction ratio.

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Abbreviations

GST	Ge₂Sb₂Te₅
SOI	Silicon-On-Insulator
IL	Insertion Loss
ER	Extinction Ratio
PCM	Phase Change Material
HRFZ	High Resistance Float Zone

Chapter 1

Overview

In this section, we go over the motivation behind doing this project and the choice of frequency and materials used.

1.1 Introduction

1.1.1 Motivation

Integrated photonic circuits are the way forward for future systems that can overcome the current limitations that electronic systems face, such as latency between processor and memory, high power consumption, and limited bandwidth. Conventional copper interconnects limit the speed, and cross-talk increases as the density of interconnects increases. Silicon photonics platform is the leading candidate for inter-chip and intra-chip communication primarily due to its low cost and high yield fabrication availability. It can take advantage of the highly developed CMOS manufacturing process used in the Silicon (Si) microelectronics industry. Silicon photonic switches are in high demand to realize Si-based photonic integrated circuits as they offer high modulation speed and picosecond switching time.

Phase change materials (PCMs) are suitable for on-chip photonic devices and integrated circuits because of their non-volatile nature. PCMs have several prominent properties such as high optical contrast between amorphous and crystalline states, the phase transition time of sub nanoseconds, compatibility with the CMOS manufacturing process, and high scalability, making them ideal candidates for chip-scale active photonic devices. PCMs like GST can be switched rapidly and repeatedly between amorphous and crystalline phases by applying short electrical/optical pulses. However, the exploration

of chalcogenide photonics has traditionally been limited to optical and infrared frequencies. A fully functional photonic system will require a range of suitable components with higher integration, miniaturization, and complex functionalities. The unique properties of GST allow for various terahertz functionalities - thermally switchable multilevel non-volatile states, electrically driven spatial light modulation, and optically controllable ultrafast resonance switching.[2] These exceptional features could be utilized in conjunction with each other and hence enable novel devices for efficient manipulation of terahertz waves.

1.1.2 Phase change material: GST

GST is a ternary compound that is made from varying proportions of germanium (Ge), antimony (Sb), and tellurium (Te). $Ge_2Sb_2Te_5$ is the most widely used stoichiometry of GST for photonic devices [2]. The phase-change material $Ge_2Sb_2Te_5$, commonly known as GST, is reported to have an optical response superior to that of the other opto-electronically attractive phase-change material vanadium dioxide (VO_2).

The active operation in the integrated photonic devices is enabled because GST exhibits good phase change behavior between its amorphous (a-GST) and crystalline (c-GST) phases. The transition between two phases of GST results in a huge difference in optical and electronic properties. This transition is rapid and repeatable. The phase change material GST exhibits a drastic change in its real as well as the imaginary part of the refractive index ($\Delta n > 1$ and $\Delta k > 1$) when switched between its two stable phases, leading to a substantial change in optical phase and amplitude within a small active length. Because of the high refractive index difference between the two phases of GST, active devices exhibit a high on-off ratio and can have broadband operation with an ultra-compact footprint.

Another attractive property of GST-based devices is the self-holding bi-stability, i.e., no continuous power is required to maintain the device state. The self-holding bistability in GST at room temperature enables a low energy-consuming platform for non-volatile photonic components.

The switching behavior of GST from a-GST to c-GST and vice-versa is achieved by Joule heating of the GST material through the application of optical or electrical pulses. These transitions occur with a transition time of sub-nanoseconds. Moreover, the fabrication of GST-based devices is also CMOS compatible. Hence, by using these characteristics of GST combined with Si waveguides, ultra-small energy-efficient and non-volatile optical switches can be realized.

A preliminary study on chalcogenide phase change material, germanium telluride (GeTe), for active terahertz control with nonvolatility has been reported. [3] However, the versatile nature of the phase change chalcogenides and their potential use in multi-dimensional terahertz wave manipulation remains largely unexplored. Traditionally, germanium antimony telluride (GST) is the popular chalcogenide phase change material that fueled the optical memory era, owing to its faster switching speed and retention time. [4] More recently, GST has led to the accomplishment of various hallmark nanophotonic devices, such as integrated all-photonic non-volatile memory,[5] optical color rendering,[6] nano pixel displays,[7] and reconfigurable nanoplasmonic devices.[8] [9]

1.1.3 Frequency of Operation: 1 THz

The Terahertz is a portion of the electromagnetic spectrum that lies between microwaves and infrared. Typically, it is loosely defined to extend from about 0.3 THz (a wavelength of 1 mm) to around 10 THz (30 m). Terahertz waves fall between the worlds of photonics and electronics. Terahertz radiation starts at a wavelength of around one millimeter and proceeds into shorter wavelengths. Hence it is also known as the submillimeter band or submillimeter waves. Terahertz waves occupy a middle ground between microwaves and infrared light waves, which has traditionally been called the “terahertz gap,” because technology for its generation and manipulation has been in its infancy compared to other regions of the EM spectrum. The generation and modulation of electromagnetic waves in this frequency range cease to be possible by the conventional electronic devices used to generate radio waves and microwaves, requiring new devices and techniques.

One thing to note is that the earth’s atmosphere is a strong absorber of terahertz radiation, so the range of terahertz radiation in the air is limited to tens of meters, making it unsuitable for long-distance communications; however, at distances of 10 meters, the band may still allow many useful applications in imaging and construction of high bandwidth wireless networking systems, especially indoor systems. Besides, producing and detecting coherent terahertz radiation remains technically challenging, though inexpensive commercial sources now exist in the 0.3–1.0 THz range (the lower part of the spectrum), including gyrotrons, backward wave oscillators, and resonant-tunneling diodes.

Terahertz radiation has many benefits over its counterpart in the spectrum & technology is becoming competitive to existing technologies in various areas like biology, spectroscopy, security, and communication. THz science and technology have advanced significantly in the last two decades, and the unique nature of THz beams has led to

their widespread and important applications in various fields. At its core, these are the properties that stand out –

Non-damaging - Electromagnetic radiation at terahertz frequencies has far less energy than electromagnetic radiation at higher frequencies (for example, x-rays). The following equation describes the energy of electromagnetic radiation:

$$E = h\nu \quad (1.1)$$

where E is the photon's energy, ν is its frequency, and h is Planck's constant. (~ 4 meV). The energy of these photons at terahertz frequencies is too low to knock electrons off atoms, i.e., they do not have the potential to ionize materials. Therefore, these waves won't damage living tissue, making them very attractive for medical uses and security applications, such as scanning airline passengers. Moreover, THz radiation penetrates cloth and leather but just a few millimeters into the skin without the potentially damaging effects of X-rays.

Higher resolution than other safe wavelengths – In general, electromagnetic radiation across the spectrum can be used to produce images. X-rays are a commonly used familiar example. The level of detail of an image depends on the wavelength of the electromagnetic waves being used. As the wavelength decreases, the resolution of the produced image increases. Terahertz radiation may not have the same level of resolving power as x-rays or visible light, but they are still better performing than microwaves or radio waves. Besides, it is not always desirable to use UV or X-rays because they are hazardous to living cells. On the other hand, visible light cannot be used to image in all circumstances, for example, when the object being imaged does not interact with visible light or is covered in something opaque to visible light. This leads on to another practical essential aspect of terahertz radiation.

Many common materials are already transparent to terahertz radiation - Many commonly used materials that are opaque to visible light (380nm to 750nm) are transparent to terahertz radiation, including many kinds of textiles, paper, and cardboard. This potentially opens up an opportunity for security applications. For instance, terahertz radiation could scan people for concealed weapons without employing the traditionally used but damaging x-rays. This technology is already in use in some airports, including Schiphol in the Netherlands.

Terahertz wavelengths correspond with many organic and biological molecules' behavior - Terahertz waves span a section of the EM spectrum at which many molecules emit and absorb radiation. Molecules have a distinct chemical “fingerprint” due to the precise

wavelength of electromagnetic radiation they emit and absorb. This can readily be used in a technique called spectroscopy to identify substances. This technique could identify concealed plastic explosives or analyze a patient's breath. Terahertz scanning also has the promising potential to identify skin cancers that are too small to observe with the naked eye. Many of the complex organic chemicals which are used in explosives absorb terahertz radiation at very particular frequencies, creating a unique "signature" that detectors can identify.

Tablet Coating - THz imaging can be quite valuable for performing 3D analysis on tablets. It can determine the coating integrity and thickness, detect and identify localized chemical or physical structures such as cracks or chemical agglomeration within a core, and interrogate embedded layers for delamination and integrity. Tablet coating has a crucial role to play in regulating the release of active ingredients in the body. Coating contributes to the bioavailability of a particular drug or combination of drugs during certain times and locations. However, the coating can also serve to protect the stomach from high concentrations of active ingredients, extend shelf life by keeping the ingredients safe from moisture and oxygen degradation, and improve tablet visual appeal.

Bandwidth and data transfer - The terahertz segment of the electromagnetic spectrum offers massive potential for high data transmission rates. The available bandwidth constrains data rates, and the terahertz frequencies offer wide bandwidths in an uncrowded part of the electromagnetic spectrum. As higher frequencies can carry more bandwidth, terahertz signals could make a super-Bluetooth that can wirelessly transfer an entire high-definition movie within a few seconds.

1.2 Software Used

The software used in the simulations is called "CST Design Environment." This powerful software contains diverse functions for simulating electromagnetic problems and is made up of six independent functional studios:

1. CST Microwave Studio,
2. CST EM Studio,
3. CST Particle Studio,
4. CST Design Studio,
5. CST PCB Studio and

6. CST Cable Studio.

CST Microwave Studio is a tool specialized for the fast and accurate 3-D ElectroMagnetic simulation of high-frequency problems. Along with a broad range of applications, CST Microwave Studio offers considerable product-to-market advantages: virtual prototyping before physical trials, shorter development cycles, and optimization instead of experimentation. All the simulations in this report were accomplished using the CST Microwave Studio. Microwave Studio is a functional module that can deal with complex high-frequency problems and has the following advantages -

Firstly, it contains the transient solver for the efficient calculation of lossy and loss-free structures. The solver performs a broadband calculation of S-parameters from one single calculation run by applying DFTs to time signals. The frequency-domain solver can process adaptive sampling and has an expert system-based automatic mesh generation with 3D adaptive mesh refinement. The general-purpose solver supports both hexahedral and tetrahedral meshes. Besides the general-purpose solver, the frequency domain solver also contains two specialized solvers for strongly resonant structures (for use with hexahedral meshes only). The first of these solvers only calculates S-parameters, whereas the second one also calculates fields and, therefore, requires additional calculation time.

Secondly, it can calculate 3D Eigenmodes. One can view the structure either as a schematic or as a 3D model. The schematic allows for easy coupling of the EM simulation with circuit simulation. It contains a fast and memory-efficient FI method. The FIT (Finite Integration Technique) method that the studio uses for simulation performs exceptionally well due to the Perfect Boundary Approximation (PBA) for solvers using hexahedral grids. The transient and Eigenmode solvers also support the Thin Sheet Technique (TST). All solvers support hexahedral grids.

Thirdly, it contains an advanced ACIS-based, parametric solid modeling front-end function with excellent structure visualization. This means that simple modeling can be directly accomplished in the studio, and a feature-based hybrid modeler allows for quick structural changes. Meanwhile, it can also import or export models of all file types; for example, SAT, Step, Autodesk Inventor, VDA-FS CATIA, Pro/E, IGES, NASTRAN, or STL 3D CAD data, and DXF 2D CAD data. It can also import GDSII and Gerber 2D CAD data.

Chapter 2

The Switch

2.1 Basic Structure of the Switch

The drive behind this work is to provide two useful transmission states for the rib waveguide by embedding GST between silicon waveguide strips. Those two self-sustaining states correspond to the amorphous and crystalline phases of GST. The SOI approach is selected to make this structure easy to manufacture and compatible with the silicon “CMOS photonics” industry.

From a theoretical standpoint, the proposed structure in Fig. 2.1 can work properly. In each phase, the GST has a complex index of refraction, and the optical waveguide performance is linked to a change in both the real and imaginary indices. We can quantify optical performance using some definitions of indices and absorption. The GST refractive indices are written for the amorphous phase as $n_{am} + ik_{am} = n_1 + ik_1$, and for the crystalline phase, $n_{cr} + ik_{cr} = n_2 + ik_2$.

Before simulations can be started, it is necessary to have reliable values of the complex indices $n + ik$ of the materials used. Literature sources of n and k have been studied, and the values that have been used for each material are mentioned in the next section. In the

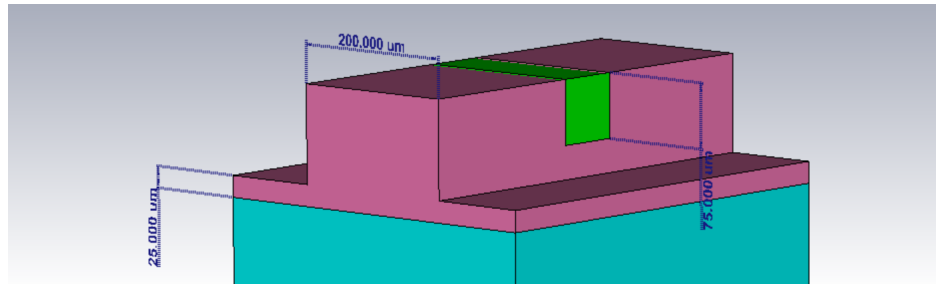


FIGURE 2.1: Model Showing dimensions of one of the geometries of the switch

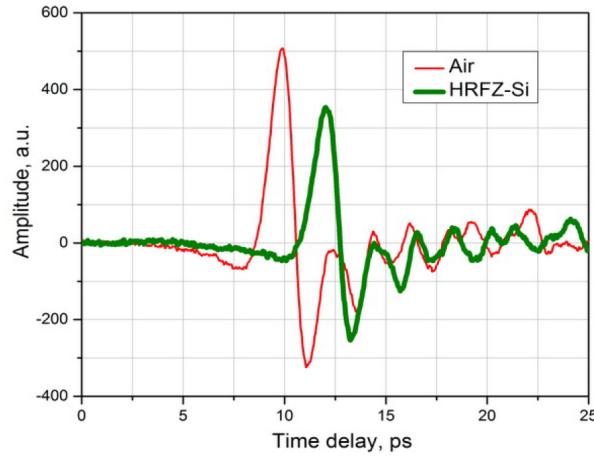


FIGURE 2.2: THz signal transmittance through air and HRFZ-Si. [adapted from [1]]

case of GST, partially crystallized or “fractional” intermediate states are not considered; only the amorphous and the rocksalt-structured crystal phases are examined.

2.2 Material Properties

2.2.1 HRFZ-Si

For THz applications, traditionally High Resistivity Float Zone Silicon (HRFZ-Si) is used, as it is the most extensively investigated substance for operating within this range. It also has a good transmission performance and besides synthetic diamond, it is the only isotropic crystalline material most suitable for the extremely wide range operation from Near Infrared ($1.2 \mu\text{m}$) to Milli-meter ($1000 \mu\text{m}$) waves. And HRFZ-Si is rather cheaper to grow and machine in comparison with diamond. HRFZ-Si can have considerably bigger dimensions, allowing manufacturing of the elements of fast-developing THz electronics.

For THz applications, High Resistivity Float Zone Silicon (HRFZ-Si) maintains 50 to 54 transmission upto $1000 \mu\text{m}$. [1] HRFZ-Si has low losses in THz range. As shown in Fig. 2.2, the THz waveform of HRFZ-Si is similar to the THz waveform of air. That indicates the lack of HRFZ-Si absorption.

Wave and free-carrier interaction is one of the fundamental kinds of wave material interactions. For THz waves interacting with conductors or semiconductors having high free-carrier density, the free-electron interaction dominates. A THz wave interacting with free carriers can be solved using the classic **Drude model** [10].

In this model, individual carriers are assumed independent of each other, and no interaction between carriers is considered except for collisions. The collision between carriers is considered an instantaneous event, and the mean interval between two collisions involving the same carrier is defined as an average collision time τ , independent of the carrier's location or velocity.

Based on these approximations, we have the following **motion equation** describing a free carrier driven by an electric wave:

$$m^* \frac{d^2x}{dt^2} + \frac{m^*}{\tau} \frac{dx}{dt} - qE = 0 \quad (2.1)$$

where m^* is the effective mass of the carrier, q denotes the carrier's charge, and E represents the electric field. Under the condition of equilibrium, the **average collision time** can be described as –

$$\tau = \frac{m^* \mu}{q} \quad (2.2)$$

Here μ is the mobility of the carrier. For instance, the effective mass of an electron in undoped silicon is $m^* = 0.19 m_0$, and its mobility is $\mu = 1,400 \text{ cm}^2/\text{Vs}$. This results in an average collision time of 1.5 ps between two electrons.

The material is polarized because of the carrier displacement induced by the electric field –

$$P = (\epsilon - \epsilon_\infty) \epsilon_0 E = Nqx \quad (2.3)$$

where ϵ_∞ is the high-frequency relative permittivity of the material, N is the free carrier density, ϵ_0 is permittivity in vacuum, and ϵ is the relative permittivity of the material at the particular frequency of the interacting electromagnetic wave.

The term “dielectric constant” of the material is often used instead of the relative permittivity, and usually the prefix of “relative” is also ignored and permittivity is used instead. Further, using the polarization of the material, the **motion equation** can be derived into a **polarization equation**:

$$\frac{d^2P}{dt^2} + \gamma \frac{dP}{dt} - \frac{Nq^2}{m} E = 0 \quad (2.4)$$

Here $\gamma = 1/\tau$ denotes the coherent decay factor of the electrons in the material.

Any electromagnetic wave can be described in the form of the sum of a series of monochromatic waves via Fourier transform. The interaction between wave and material can also be presented as the sum of material interacts with the series of monochromatic waves. Each monochromatic wave and its induced polarization are simple harmonic oscillations. Solving the above equation results in the complex permittivity of the material at a particular frequency, $\epsilon^* \equiv \epsilon_1 + i\epsilon_2$

$$\begin{aligned}\epsilon(\omega) &= \epsilon_\infty - \frac{\epsilon_\infty \omega_p^2}{\omega^2 + i\omega\gamma} \\ &= \epsilon_\infty \left(1 - \frac{\omega_p^2}{\omega^2 + \gamma^2} + i \frac{\omega_p^2 \gamma}{\omega(\omega^2 + \gamma^2)} \right)\end{aligned}$$

with

$$\omega_p = \sqrt{\frac{Ne^2}{m^* \epsilon_\infty \epsilon_0}} \quad (2.5)$$

which is called the plasma oscillation frequency (POF) of the material. The Plasma Oscillation Frequency (POF) is proportional to the square root of the free-carrier density in the material. For example, metals have very high electron density, so that they have high POF located in the UV band. The free-carrier density in semiconductors varies with different materials and environmental conditions, such as doping, temperature, etc. Therefore, semiconductors have a variety of POFs.

Crystalline silicon has a high-frequency permittivity of $\epsilon_\infty = 11.7$. For intrinsic silicon, whose free-electron density is only in the order of 10^{10} cm^{-3} , ω_p is less than 1/100 of THz. As a result, a THz wave is transparent in intrinsic silicon. Hence, the calculated values of dielectric constants at 1 THz using $\epsilon_\infty = 11.7$, $\gamma = (1.5\text{ps})^{-1}$ and $\omega_p = 0.01$ THz is -

$$\epsilon = 11.69 + i(0.0005)$$

The graph in Fig. 2.3 shows the permittivity of silicon at 1 THz with different doping densities

The complex dielectric constant of silicon depends on its conductivity (i.e., free-carrier concentration). For a low impurity concentration, the dielectric permittivity is almost a real value, approximately equal to the high frequency dielectric permittivity. As impurity concentration increases, the real part of the dielectric constant becomes a negative value, and its imaginary part cannot be considered negligible. The dielectric permittivity, therefore, presents its complex nature, and silicon becomes lost to THz wave.

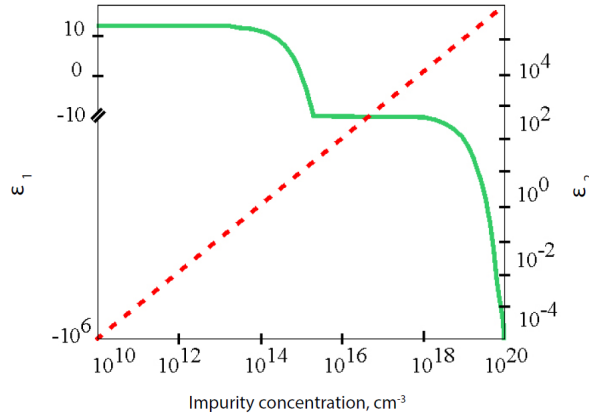


FIGURE 2.3: Real (solid) and imaginary (dashed) part of dielectric permittivity of n-type silicon with changing impurity concentration at 1 THz [adapted from [1]]

2.2.2 Quartz (SiO_2)

One of the best materials for wavelengths greater than $50 \mu\text{m}$ is z-cut crystal quartz. The z-cut crystal quartz windows are transparent in the visible range, making it easy to adjust the optical system through HeNe laser. They do not change the light polarization state and can be cooled down below the λ -point of liquid helium.

For optical characterization at terahertz frequencies, Quartz is commonly used owing to its high transmission and low absorption at these frequencies. Quartz shows outstanding transmission and is thus an ideal optical substrate over the THz band, apart from the region $3.9 \pm 0.1 \text{ THz}$ owing to a spectral feature originating from the lowest energy optical phonon modes.

Quartz is an earth-abundant mineral form of tetrahedrally bonded silica (i.e., silicon dioxide, SiO_2). At temperatures below 846 K, the stable crystalline form of Quartz is the α -phase which is uniaxial and has a chiral structure that may be right or left-handed [11]. Thus light propagating along the optic axis (also called c-axis or z-axis) suffers no birefringence but experiences optical activity, while light propagating along other directions exhibits birefringence.

At terahertz frequencies, x-cut Quartz has been shown to be highly birefringent over the range 100 GHz to 3.5 THz [12], and stacks of birefringent quartz plates have been used to create an achromatic quarter waveplate operating at terahertz frequencies [13].

When used as substrates for optical and THz spectroscopy, wafers of z-cut Quartz are typically used to avoid birefringence affecting light propagating normal to the surface of the wafer, which is the usual measurement geometry.

Frequency	n_0	α_0 (cm^{-1})
0.90	2.1076	0.10
1.05	2.1083	0.10

TABLE 2.1: Quartz refractive index & absorption coefficient

Moreover, Quartz is opaque throughout most of the 2000-250 cm^{-1} spectral range in thicknesses greater than a few tenths of a millimeter, and, therefore, it is particularly beneficial for eliminating high-order radiation in far-infrared grating spectrometers. Experimental values of refractive index for Quartz have been used from ref. [14] and are mentioned in Table 2.1.

Using the relation $\alpha = 4\pi\nu\kappa$, the value of κ has been calculated. The following relations were then used to find out the dielectric values of Quartz –

$$n^* = n + i\kappa$$

$$\epsilon_r^* = \epsilon_r' + i\epsilon_r''$$

$$(n^*)^2 = \epsilon_r^*$$

Crystalline Quartz satisfies both the criteria, having a relatively low permittivity and comparatively low absorption losses at THz frequencies, and hence, is the substrate of choice.

2.2.3 GST: $\text{Ge}_2\text{Sb}_2\text{Te}_5$

The applicability of GST for various photonic applications comes from its three crystallographic phases – amorphous, metastable FCC (face-centered cubic), and stable HCP (hexagonal close-packed) lattice structures, which possess strikingly different electronic and optical properties.

The crystallization temperature of GST to FCC phase has been reported as ≈ 150 °C and to HCP phase as ≈ 250 °C. [15]. It is important to emphasize that these states of GST are non-volatile with zero hold power and are expected to have a retention time of years. [16]

$\text{Ge}_2\text{Sb}_2\text{Te}_5$ provides a means of achieving a non-volatile modulation based on the base terahertz conductivity through thermal annealing, as well as an ultrafast volatile modulation over the base modulation through optical pumping. [2]. The extracted values of

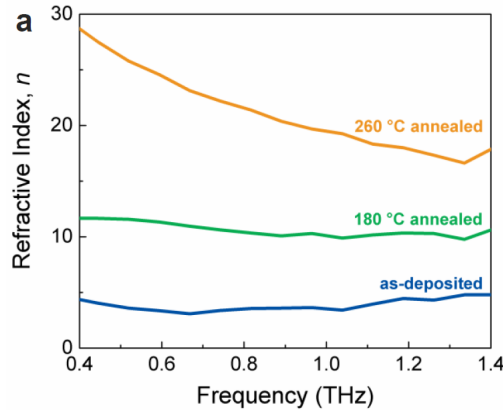


FIGURE 2.4: Refractive Index as a function of frequency for different phases of GST [adapted from [2]]

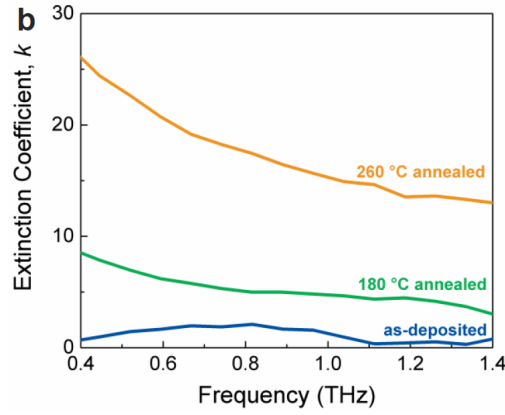


FIGURE 2.5: Extinction Coefficient as a function of frequency for different phases of GST [adapted from [2]]

(i) refractive index, n and (ii) extinction coefficient, k for amorphous, 180 °C annealed, and 260 °C annealed GST films, respectively, are shown in Figure 2.4 & Figure 2.5 (adapted from [2]).

The refractive index of as-deposited amorphous GST thin film at 0.6 THz was measured to be ~ 4 and which switches to 12 and 25 when annealed at 180 °C and 260 °C, respectively. Such a massive contrast in refractive index is especially attractive for frequency-agile terahertz devices, where the resonance frequency is spectrally shifted without the modulation of the resonance strength.

However, the large change in the refractive index is also accompanied by a significant increase in the extinction coefficient (with an increase in crystallinity) and leads to higher terahertz absorption in the GST film. Hence, GST can be used to realize the two major branches of terahertz devices - frequency agile and resonance modulation devices.

Chapter 3

Simulations

3.1 Designing the waveguide

The initial task before designing the switch was to model simulate a working Silicon-on-insulator based integrated rectangular rib waveguide for operation at THz frequencies and optimize for frequencies centered around 1 THz. The simulations were initially carried out for cross-section height x width = $50 \times 200 \mu m^2$ of the Si waveguide, which supports a single mode at these dimensions [17]. The height of the proposed waveguide was then varied from $25 \mu m$ to $250 \mu m$ in steps of $25 \mu m$. The width of the waveguide was varied from $200 \mu m$ to $350 \mu m$. The length for the waveguide was kept fixed at 1 mm with a rib height of $25 \mu m$ for all the simulations. The 3D plot shown in fig 3.1 was then obtained for the transmission loss of the waveguide.

As can be clearly seen from Fig. 3.1, the loss [in dB] varies from - 0.024 dB to - 0.047 dB. This translates to a transmission loss of 0.24 dB/cm to 0.47 dB/cm. The reported transmission losses in literature for different waveguides are 4.3 dB/cm (HDPE), 1.35 – 1.92 dB/cm (rectangular hollow metallic waveguides), 3 dB/cm (hollow core circular waveguides) [17].

For our purpose of designing a switch, the change in propagation loss with the varying dimensions of the dielectric Si rib waveguide is negligible. Hence, to minimize the cross-section and reduce the footprint of the switch, an optimized cross-section of **$200 \mu m$ x $114 \mu m$ (width x height)** was chosen for further simulations.

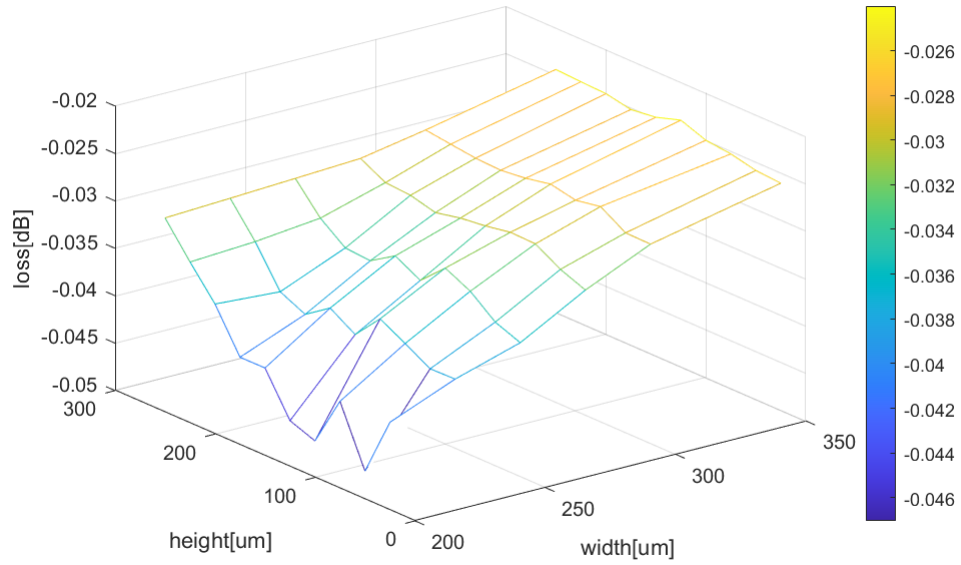
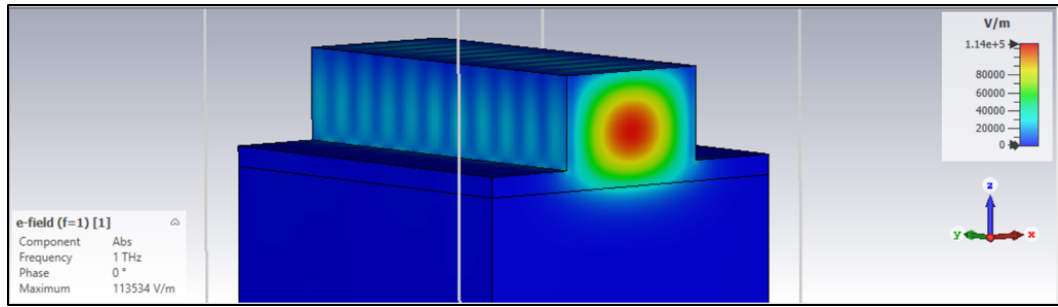


FIGURE 3.1: Transmission loss of the proposed rib waveguide (in dB)

FIGURE 3.2: Mode Profile of the proposed waveguide of cross-section $200\mu\text{m} \times 114\mu\text{m}$ (width x height) (as simulated)

3.2 Designing the switch

For designing a GST-based photonic switch, we take the approach of embedding GST inside the etched silicon-on-insulator (SOI) waveguide. This approach improves the light-matter interaction and provides a high extinction ratio within a small active volume. The phase-change material GST embedded into the Si strip waveguide interacts with the propagating mode depending upon the phase of this material.

To incorporate GST, two suitable methods have been employed – 1.) Fully etched Si, and 2.) Partially etched Si.

In the **first** method (3.3), Si is removed entirely up to the rib layer, and GST is deposited in this space. The remaining height is left uncovered. In the **second** approach (3.4), the Si is removed partially in the middle of the waveguide & GST is deposited in this space to match the height of the actual waveguide. In the second approach, H_{GST} and H_{Si} are

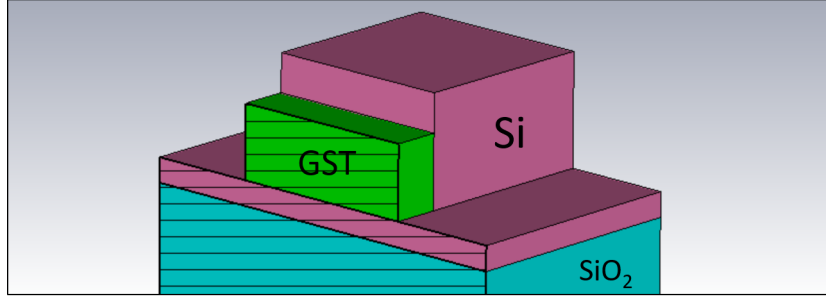


FIGURE 3.3: Cross-section view of the fully etched switch

varied so that the total height of the waveguide remains the same ($H_{GST} + H_{Si} = 114 \mu\text{m}$).

Hence, the light launched from the input port traverses some length initially in the Si waveguide, interacts with GST, and finally traverses a length again in the Si waveguide before reaching the output port. Because of the different values of the GST extinction coefficient, the optical mode suffers variable attenuation in two phases.

In the crystalline phase, because of the considerably large value of the extinction coefficient, GST behaves as a lossy waveguide, and the propagating mode suffers significant attenuation as most of the light is absorbed by it. In contrast, the mode propagates with lesser attenuation in the amorphous phase because of the small value of the extinction coefficient than that of the crystalline phase. Therefore, the amorphous and **crystalline** phases of GST correspond to the ON and **OFF** states of the photonic switch, respectively. Considering the insertion loss and the extinction ratio, we need to find the optimized value of GST dimensions (length, width & height).

3.2.1 Performance Parameters

The switch designed here is a hybrid Si-GST one input-one output (1×1) waveguide switch. The Si rib waveguide is designed in an SOI platform with a $25 \mu\text{m}$ (height) Si rib layer and a $500 \mu\text{m}$ thick buried silicon dioxide substrate (which can be varied).

For single-mode operation at 1 THz ($300 \mu\text{m}$), the cross-section of the waveguide is chosen $200 \mu\text{m} \times 114 \mu\text{m}$ (width $\times$ height). The output optical power through the Si waveguide is altered by changing the phase of GST (from a-GST to c-GST and vice-versa) embedded inside the Si waveguide. The optical mode is launched from the left side, and the output is measured from the right side of the waveguide (3.3). To achieve a high extinction ratio and low insertion loss, the GST dimensions are varied.

To measure the performance, Insertion Loss & Extinction Ratio are defined as -

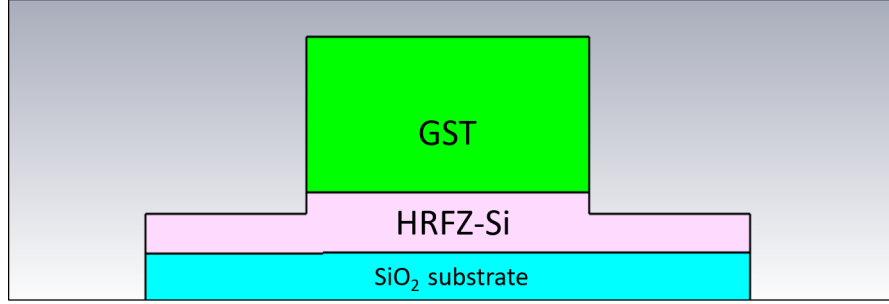


FIGURE 3.4: Front cross section view of the partially etched switch

TABLE 3.1: Values of refractive indices used for optical simulations

Material	Refractive Index ($n+ik$)	Epsilon (Real)	Epsilon (Imaginary)
a-GST	$3.66 + i(1.383)$	$\epsilon' = 11.645271$	$\epsilon'' = 9.68436$
c-GST	$19.56 + i(15.331)$	$\epsilon' = 147.554$	$\epsilon'' = 599.748$
$\text{SiO}_2(\text{Quartz})$	$2.10 + i(0.0002)$	$\epsilon' = 4.4434$	$\epsilon'' = 0.00101$
HRFZ-Si	3.42	$\epsilon' = 11.699$	$\epsilon'' = 0.00054$

$$\text{IL} = 10\log(T_{ON}),$$

$$\text{ER} = 10\log(T_{ON}/T_{OFF}),$$

where T_{ON} and T_{OFF} denote the transmission for the amorphous and crystalline phases, respectively.

3.3 Optical Simulations

Optical simulations are performed using the optical module of CST Microwave Studio. This software uses the finite integration method, and perfectly matched layer boundary conditions are considered for simulations.

The optical properties of the device are modeled, taking the refractive index of SiO_2 , Si, and Air as 2.1, 3.42, and 1, respectively, at 1 THz. For the simulations performed, we define the refractive indices as $n + ik$, where n and k denote the real and imaginary parts of the material's refractive index, respectively. For GST, we use the value of refractive index $n_{am} + i_{am} = 3.66 + i1.323$ (a-GST) and $n_{cr} + i_{cr} = 19.56 + i15.331$ (c-GST) at 1 THz, where n is the real part and k is the imaginary part of the refractive index. All the values of the refractive index used are summarized in Table 3.1 –

Here epsilon is related to the values of refractive index $n + ik$ as –

$$\epsilon' = n^2 + k^2,$$

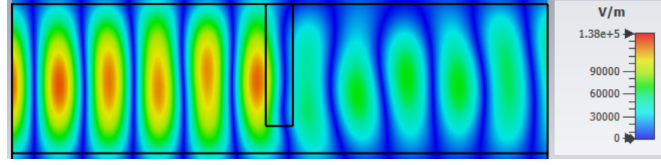


FIGURE 3.5: Demonstration of the ON state (a-GST)

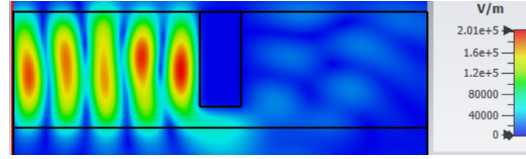


FIGURE 3.6: Demonstration of the OFF state (c-GST)

$$\epsilon'' = 2nk$$

To demonstrate the working of the designed photonic switch, a representational plot for the absolute value of the electric field profile on the partially etched Si waveguide has been shown in Fig. 3.5. The refractive index contrast between the crystalline and amorphous phases of GST results in different electric field profiles of the propagating mode and, therefore, different output powers at the end of the waveguide.

In the **amorphous phase**, Fig. 3.5, the mode propagates through GST without any appreciable attenuation because of the small value of the extinction coefficient, and thus with high transmission, depicts the **ON** state of the switch.

In contrast, the **crystalline phase**, Fig. 3.6, is highly lossy, and the mode passing through GST suffers significant attenuation due to high absorption. Therefore, with the almost negligible transmission, the crystalline phase depicts the **OFF** state of the switch.

Chapter 4

Observations

To numerically investigate the dependence of switch parameters IL and ER on the dimensions of GST film, simulations are performed for both amorphous and crystalline phases for various physical configurations (Fig. 4.1) of the switch.

4.1 Set 1

Results of the first set of simulations that correspond to Fig. 4.1(a) are shown in Fig. 4.2. In this case, the embedded GST length, L_{GST} , is varied from $50\text{ }\mu\text{m}$ to $150\text{ }\mu\text{m}$ along

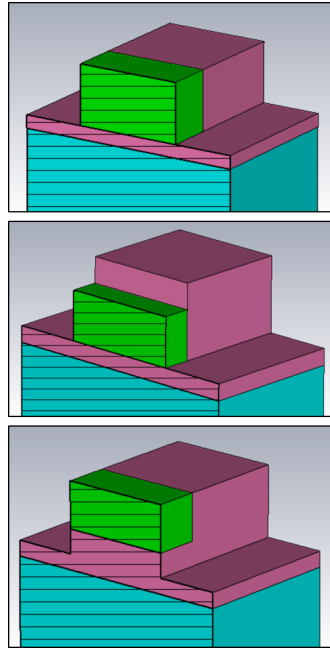


FIGURE 4.1: Schematic view of the proposed switch (a) fully etched Si waveguide with GST cross-section matching Si cross-section, (b) fully etched Si waveguide with GST height variant, and (c) partially etched Si waveguide partially filled with GST

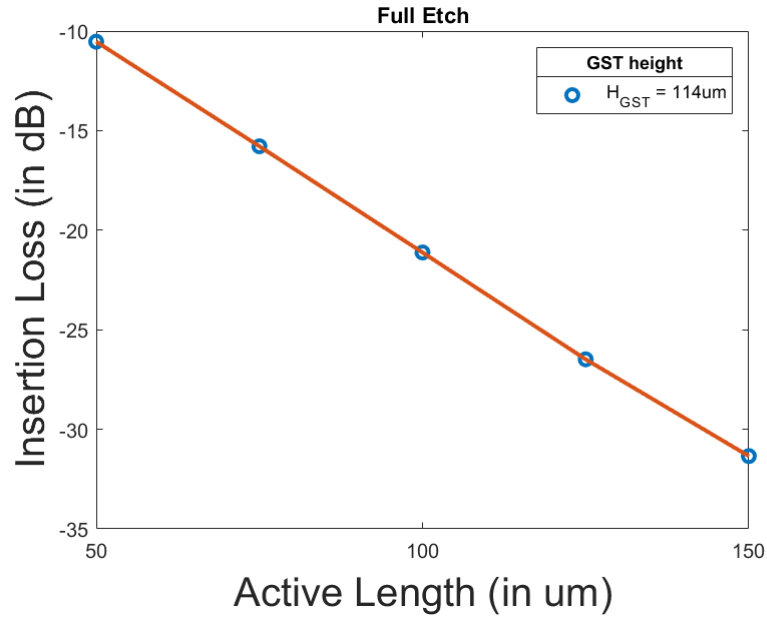


FIGURE 4.2: Variation of IL with active length of GST for configuration 1

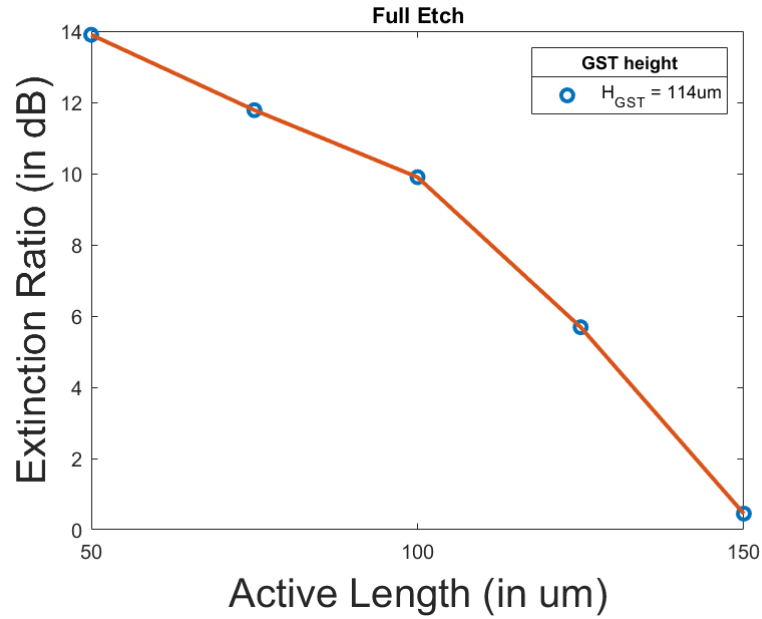


FIGURE 4.3: Variation of ER with active length of GST for configuration 1

the direction of mode propagation at intervals of $25 \mu\text{m}$ while keeping the cross-section the same as that of Si waveguide. When the light propagating in the Si waveguide reaches the etched part filled with GST, losses occur because of GST absorption and reflections at Si-GST interfaces. These losses are higher in crystalline GST (c-GST) than those in amorphous GST (a-GST). This is because both imaginary and real parts of the refractive index are much higher for c-GST than those for a-GST. We observe the highest **ER of 13.89 dB** at $L_{GST} = 50 \mu\text{m}$ at the expense of an **IL of 10.53 dB**.

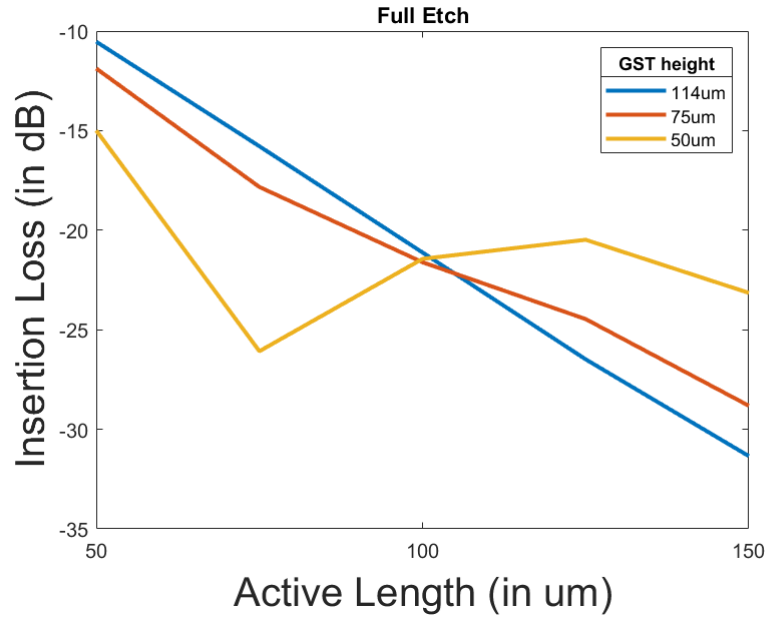


FIGURE 4.4: Variation of IL with active length of GST for configuration 2

4.2 Set 2

In the second configuration, the Si waveguide is entirely etched, and GST is embedded in the channel with variation in height H_{GST} (Fig. 4.1(b)) from 50 to 114 μm at different values of L_{GST} from 50 to 150 μm .

Results of the second set of simulations are shown in Fig. 4.4. These simulations exhibit the dependence of ER and IL on the GST height. Fig. 4.5 illustrates the ER trend. As shown in Fig. 4.5, the peaks in the ER curve are due to variable reflection, scattering, and absorption for different values of H_{GST} and L_{GST} combination in c-GST. For c-GST, the amount of light absorbed, reflected, and scattered by the GST layer for certain combinations of H_{GST} and L_{GST} is greater than that for other combinations of H_{GST} and L_{GST} . A maximum **ER** of ≈ 18.48 dB is obtained at $L_{GST} = 100$ μm and $H_{GST} = 75$ μm with an **IL** of 21.61 dB.

4.3 Set 3

We partially etched a Si waveguide in the third case and filled the remaining space with GST, as shown in Fig. 4.1(c). Parameter sweep was performed by varying the heights H_{GST} , and H_{Si} from 50 to 100 μm at different values of L_{GST} , and H_{GST} is complementary with H_{Si} and vice versa.

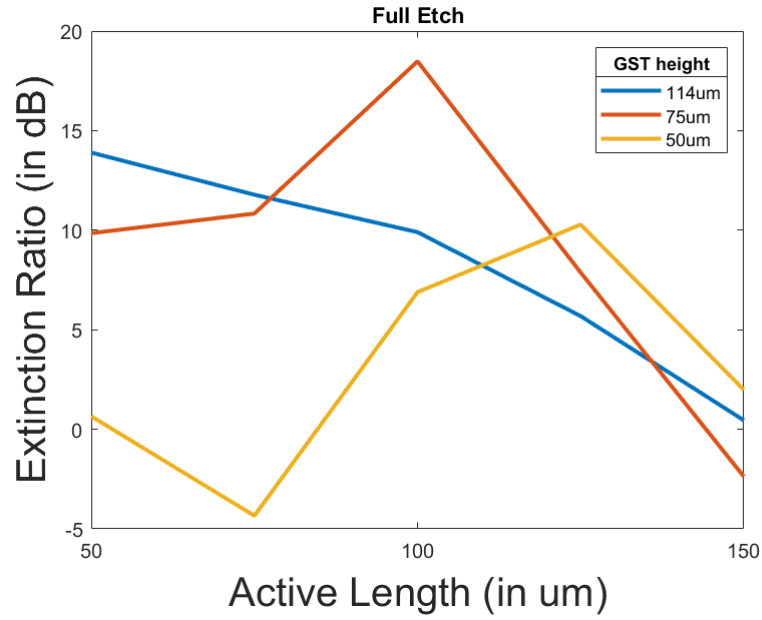


FIGURE 4.5: Variation of ER with active length of GST for configuration 2

Results for this set of simulations are shown in Fig. 4.6 & Fig. 4.7. Results in Fig. 4.7 show the ER trend, and a **maximum ER of 8 dB** is observed at $H_{GST} = 100 \mu\text{m}$ and $L_{GST} = 50 \mu\text{m}$ with an **IL of 8 dB**.

When light launched from the input port of the Si waveguide reaches the hybrid waveguide part (consisting of partially etched Si and filled with GST), it is divided between the GST layer and Si layer. The amount of light passing through the GST layer (and Si) varies with the values of H_{GST} and H_{Si} . In the crystalline phase, the amount of light absorbed, reflected, and scattered by the GST layer is greater for certain combinations of H_{GST} and H_{Si} than that for other combinations of H_{GST} and H_{Si} .

4.4 FOM: Figure-of-Merit

The photonic switches designed here are non-volatile, and no electrical bias is needed to maintain a particular state (0 or 1). The relevant parameters for static performance are ER, IL, and active volume for such photonic switches. The electrical bias is not relevant here.

Therefore, to have a comprehensive idea of the designed photonic switches, we propose the figure-of-merit (FOM) defined as

$$\text{FOM} = \text{ER} / (\text{IL} \times \text{active volume})$$

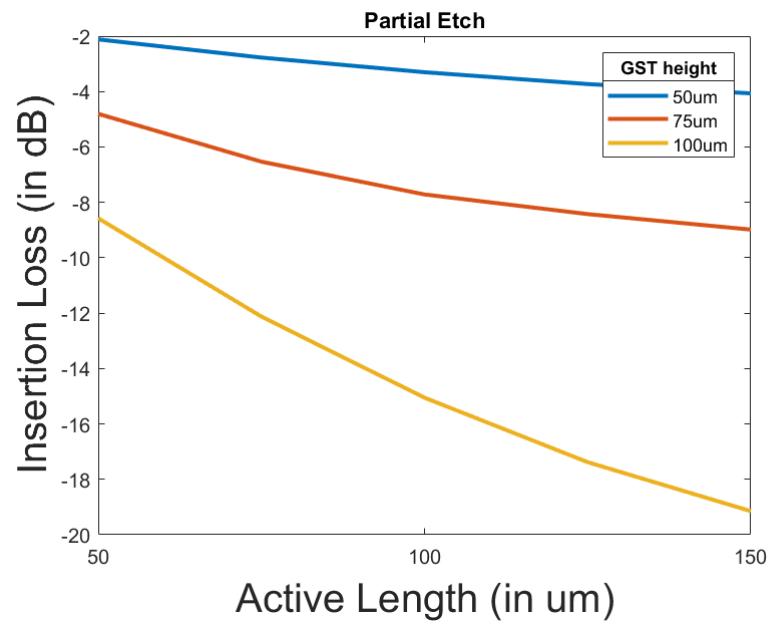


FIGURE 4.6: Variation of IL with active length of GST for configuration 3

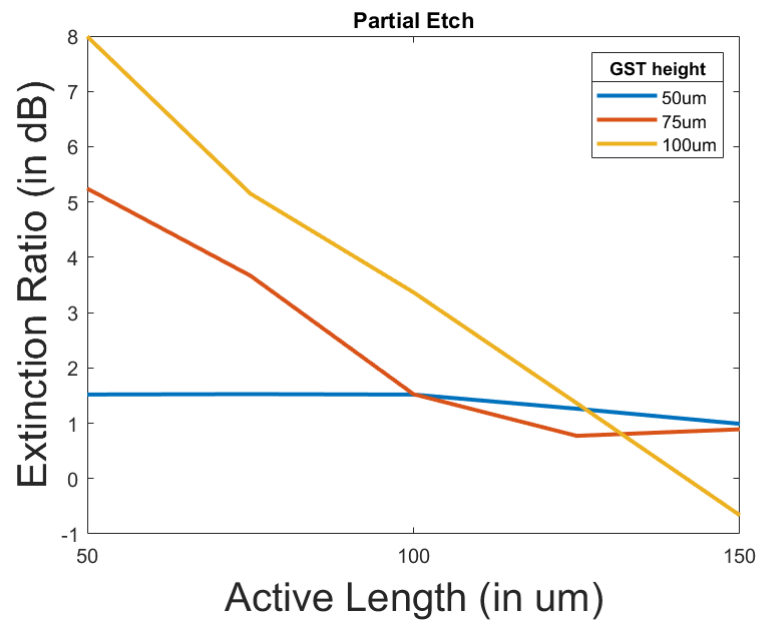


FIGURE 4.7: Variation of ER with active length of GST for configuration 3

TABLE 4.1: FOM values corresponding to the highest ER in each case

Design	ER (in dB)	IL (in dB)	Active Volume ($\times 10^6 \mu m^3$)	FOM ($\times 10^{-7} \mu m^{-3}$)
Fig. 4.1a	13.89	10.53	1.14	11.570
Fig. 4.1b	18.48	21.61	1.5	5.701
Fig. 4.1c	7.99	8.57	1.0	9.319

where ER and IL are considered linear. The FOM values corresponding to the highest ER of each configuration are listed in Table 4.1.

Chapter 5

Outlook & Conclusion

In conclusion, we have designed and theoretically analyzed a non-resonant, hybrid Si-GST integrated photonic switch for static performance in the intersection world of electronics and photonics at a frequency of 1 THz.

We used three different approaches to optimize embedded GST to obtain a high ER and a low IL. For each case, the FOM values were calculated corresponding to the highest ER. The values of IL & ER are expected to change slightly when metal electrodes are deposited on GST films for the switching effect to occur.

The results of this study reveal that variation in dimensions of GST embedded into the Si waveguide leads to variations in the performance of the photonic switch. However, such high values of IL (as reported in the results) are undesirable when it comes to minimizing the overall power consumption of the device. This is due to the combined effect of absorption in GST and reflection and scattering at the interfaces of GST and Si. Hence, more rigorous simulations need to be performed to check if a high ER with correspondingly low IL can be achieved.

Hence, two possible directions to move forward are - **1.) Decreasing the active length** even further ($< 50 \mu\text{m}$), such that, there is enough attenuation in cGST to achieve a good contrast in the transmission losses of a-GST & c-GST so that we can achieve a reasonable Extinction Ratio with a minimal Insertion Loss. Preliminary simulations (one each for a-GST & c-GST) carried out for this case, with the already used cross-section of $200 \mu\text{m} \times 114 \mu\text{m}$ (width x height), but an active length of just $2 \mu\text{m}$ gives a very low IL of 0.4514 dB along with an ER of 10.24 dB. **2.) Increasing the cross-section** of the waveguide to increase the interaction of the propagating mode with GST within a small active length. Infact, preliminary simulations carried out (similar to the configuration shown in Fig. 4.1a) with GST completely filled in a fully etched waveguide

of new cross-section $350\ \mu\text{m} \times 250\ \mu\text{m}$ (width $\times$ height) gives an ER of 29.2083 dB with a corresponding IL of 11.74 dB for an active length of $50\ \mu\text{m}$. (Fig. [A.2](#))

Furthermore, the spectral response of this switch geometry can also be analyzed in continuation of this work.

In summary, this work attempts to lay the foundation for realizing PCM-based hybrid integrated electro-optic switches, photonic memories, and electro-optic modulators on a silicon platform for inter and intra-chip THz applications. Due to fabrication errors, there is a likelihood that the length after fabrication differs from the intended one. The given results should hold good to neglect those errors. However, dedicated experimental investigations are needed to verify these results. The work reported here will be helpful in the experimental realization of PCM-based hybrid integrated switches that can be built using GST on a silicon platform for application in integrated circuits for inter-chip and intra-chip photonic applications at Terahertz frequencies.

Appendix A

Simulations carried out (similar to the configuration shown in Fig. 4.1a) with GST completely filled in a fully etched waveguide of cross-section $350\text{ }\mu\text{m} \times 250\text{ }\mu\text{m}$ (width x height).

Maximum ER of 29.2083 dB with a corresponding IL of 11.74 dB is obtained for an active length of $50\text{ }\mu\text{m}$.

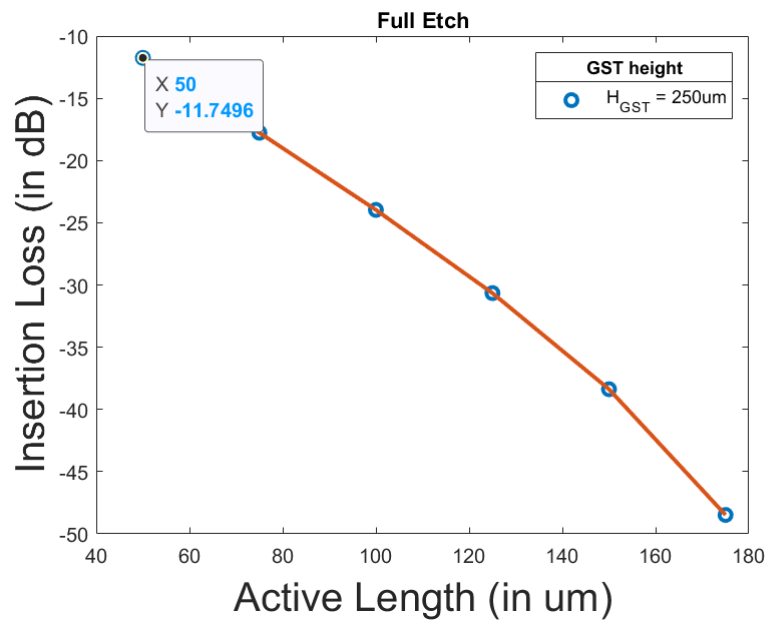


FIGURE A.1: Variation of IL with active length of GST

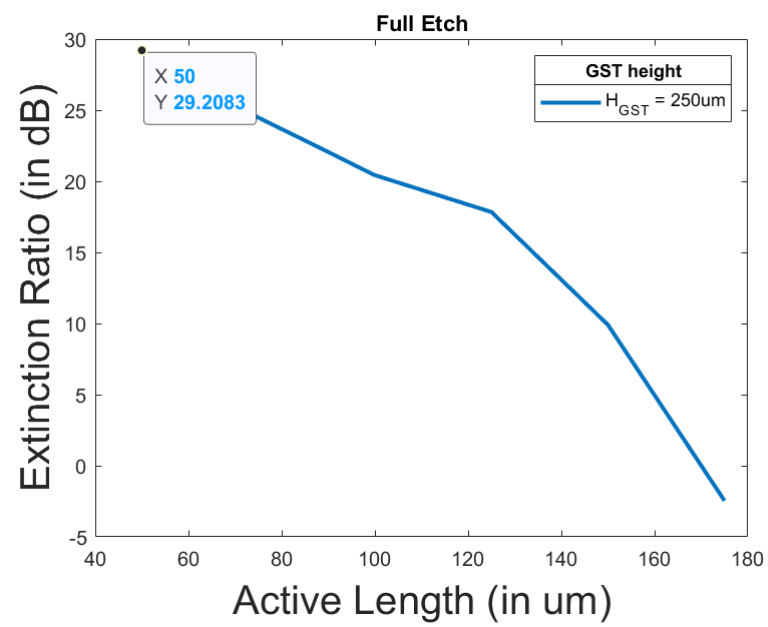


FIGURE A.2: Variation of ER with active length of GST

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